

HARIS WAHEED BHATTI

BIM Developer | AEC Software Engineer

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Seeking a full-time contract; can begin immediately under Slovenian student employment (to 30 Sep 2026).

PROFILE

Civil engineer with a European Master's in Building Information Modelling who builds software for the AEC industry – IFC schemas, graph databases and LLM agent systems. Author of `ifctoneo4j`, an open-source IFC-to-Neo4j converter published on PyPI.

PROJECTS

ifctoneo4j – Python package (PyPI) | Python, IfcOpenShell, Neo4j, Cypher 2026

- Open-source library converting IFC models (IFC2X3, IFC4, IFC4X3) into Neo4j property graphs aligned to Linked Building Data ontologies, with no RDF intermediate. Published to PyPI: `pip install ifctoneo4j`.

BIMGraphExplorer – LLM agent for IFC question answering | Python, MCP, Neo4j 2026

- Adaptive querying system exposing four read-only tools over the Model Context Protocol; profiles the graph schema and issues iterative Cypher queries conditioned on prior results.
- Raised answer accuracy by 33 percentage points and source transparency from 12.5% to 99.7% on a fixed model and dataset, through prompt design alone.
- Benchmarked against a published direct-IFC baseline under matched conditions (same model, same validated judge; 23 IFC datasets, 367 questions, 15 projects). Overall accuracy 72.0% against the baseline's 82.3%, but reached source-transparency parity (99.7% versus 100%) and outperformed it on 5 of 23 datasets, by up to 24 points on one.

Federated BIM Model – Pakistani-Themed Expo, Guimarães | IFC, Autodesk Construction Cloud Oct 2025

- Owned the federated model and clash-detection workflow across architecture and MEP disciplines, validated against Employer's Information Requirements under ISO 19650.

Blockchain-Based BIM Document Management | Solidity, Ethereum (Arbitrum Sepolia) 2024

- Smart contract system to sign and manage construction documents, with immutable on-chain linkage, to streamline dispute resolution under FIDIC contracts. Funded by a competitive grant from the Pakistan Engineering Council.

EDUCATION

MSc, European Master in BIM (BIM A+) | UL FGG, Ljubljana, Slovenia Oct 2025 – Jul 2026

- Field of study: Information Management in Construction. Completed the accelerated thesis track, finishing full MSc requirements in 10 months. Final grade 9.27/10.
- Thesis: *Graph-Mediated LLM Exploration for IFC Question Answering: A Benchmark Comparison Against Direct-IFC Approaches*. Supervisor: Prof. Žiga Turk. hdl.handle.net/20.500.12556/RUL-185169
- Self-funded living costs through part-time employment throughout.

BE Civil Engineering | NUST, Islamabad, Pakistan Sep 2020 – Jun 2024

EXPERIENCE

Contract Management Intern | Engineering and Management Services, Pakistan 2023

- Reviewed contract reports and arbitration documentation under FIDIC conditions; presented findings to the internal legal team.

Civil Engineering Intern | Fatima Fertilizer Company, Pakistan 2023

- Modelled a 6,804 m² rock phosphate storage shed in 3D from 2D drawings, coordinating with stakeholders on design refinement.

Research Intern (remote) | Southern Illinois University Edwardsville, USA 2023

- Synthesised academic literature to support a faculty-mentored research proposal.

SKILLS

Programming & Data: Python, TypeScript, Neo4j, Cypher, knowledge graphs, Git

AI / LLM: Agentic systems, Model Context Protocol (MCP), RAG, LLM evaluation and benchmarking

BIM & AEC: IFC (IFC2X3, IFC4, IFC4X3), IfcOpenShell, Linked Building Data ontologies, ISO 19650, Revit, Dynamo, Blender, Autodesk Construction Cloud, Three.js, web-ifc

Languages: English (fluent), Slovene (A1, actively learning)